



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

PRE - SCORE

15-10-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS, TNAS, TAAS & TTAS)

ANSWER KEY

PAPER-1 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,C	A,C	B	A,B,D	C	B,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	B	C	D	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	0.08	22.06	0.30	342.00	2.66 to 2.67	1.81	55.55 to 55.56	43.75

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	C,D	C,D	A,C	A,C	A,B,D	B,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	B	B	D	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	9.00	9.00	3.78 to 3.79	15.00	78.00	193.00	20.00	12.70

PART-3 : MATHEMATICS

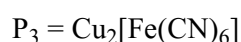
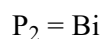
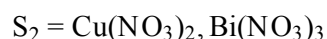
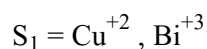
SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B	A,B,C,D	B,C	B,D	B,C	A,C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	D	C	B	D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	0.97	2.00	150.00	5050.00	8.00	17.00	8.00	8.50

HINT – SHEET

PART-2 : CHEMISTRY

SECTION-I (ii)

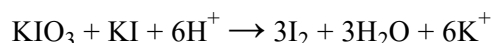
10. Ans (A)



PART-2 : CHEMISTRY

SECTION-II

3. Ans (3.78 to 3.79)

3 moles of KIO_3 = mole of H_2O

$$\Rightarrow 3 \times \frac{15}{(127 + 48 + 39)} = \frac{X}{18}$$

$$\Rightarrow X = 3.785$$

4. Ans (15.00)

E = Sulphur,

P = $S_2O_3^{2-}$

Q = S^{2-}

R = SO_2

S = H_2S

T = SO_4^{2-}

U = $S_4O_6^{2-}$

5. Ans (78.00)

$M_1 = Fe$ (Z = 26)

$M_2 = V$ (Z = 23)

$M_3 = Cu$ (Z = 29)

6. Ans (193.00)

X = NO_2^+ , NO_3^-

Y = $NaNO_3$

7. Ans (20.00)

Z = I_2Cl_6

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,B)

$$2x^3 + 4x^2 - 5x + 8 = 2(x - \alpha)(x - \beta)(x - \gamma)$$

Put $x = 2$, we get

$$30 = 2(2 - \alpha)(2 - \beta)(2 - \gamma)$$

Put $x = 1$, we get

$$9 = 2(1 - \alpha)(1 - \beta)(1 - \gamma)$$

Put $x = -1$

$$15 = 2(-1 - \alpha)(-1 - \beta)(-1 - \gamma)$$

Now multiply all,

$$(2 - \alpha)(2 - \beta)(2 - \gamma)(1 - \alpha^2)(1 - \beta^2)(1 - \gamma^2) = 15^2 \times \frac{9}{4}$$

(Option A is correct)

$$\text{Let } 1 + \alpha = x \Rightarrow \alpha = x - 1$$

$$2\alpha^3 + 4\alpha^2 - 5\alpha + 8 = 0$$

$$\text{Put } \alpha = x - 1$$

$$\Rightarrow 2(x - 1)^3 + 4(x - 1)^2 - 5(x - 1) + 8 = 0$$

$$\Rightarrow 2(x^3 - 3x^2 + 3x - 1) + 4(x^2 - 2x + 1) - 5x + 5 + 8 = 0$$

$$\Rightarrow 2x^3 - 6x^2 + 6x - 2 + 4x^2 - 8x + 4 - 5x + 13 = 0$$

$$\Rightarrow 2x^3 - 2x^2 - 7x + 15 = 0$$

2. Ans (A,B,C,D)

$$\ell = \left(1 + \frac{2}{3} + \frac{6}{3^2} + \frac{10}{3^3} + \dots \right)^{\log_{0.25} \left(\frac{1}{3} + \frac{1}{3^2} + \dots \right)}$$

$$S = 1 + \frac{2}{3} + \frac{6}{3^2} + \frac{10}{3^3} + \dots$$

$$\frac{S}{3} = \frac{1}{3} + \frac{2}{3^2} + \frac{6}{3^3} + \dots$$

$$\frac{2S}{3} = 1 + \frac{1}{3} + \frac{4}{3^2} + \frac{4}{3^3} + \dots$$

$$\frac{2S}{3} = \frac{4}{3} + \frac{4}{3^2} + \frac{4}{3^3} + \dots$$

$$S = \frac{3}{2} \left(\frac{4/3}{1 - 1/3} \right) = 3$$

$$\text{Now } \ell = (3)^{\log_{0.25} \left(\frac{1/3}{1 - 1/3} \right)}$$

$$\ell = 3^{\log_{(1/4)} \left(\frac{1}{2} \right)} = 3^{1/2} = \sqrt{3}$$

$$\Rightarrow \ell^2 = 3$$

3. Ans (B,C)

$$\sum_{r=1}^{2023} f(r) = 1 + 1 + \frac{1}{2} + 1 + \frac{1}{2} + \frac{1}{3} + \dots + 1 + \frac{1}{2} + \dots + \frac{1}{2023}$$

$$= 2023 \times 1 + 2022 \times \frac{1}{2} + 2021 \times \frac{1}{3} + \dots +$$

$$\frac{1}{2023}$$

$$= 2023 \times 1 + (2023 - 1) \frac{1}{2} + (2023 - 2) \frac{1}{3} + \dots + (2023 - 2022) \frac{1}{2023}$$

$$= 2023 \left(1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{2023} \right) - \left(\frac{1}{2} + \frac{2}{3} + \dots + \frac{2022}{2023} \right)$$

$$= 2023f(2023) - \left[1 - \frac{1}{2} + 1 - \frac{1}{3} + \dots + 1 - \frac{1}{2023} \right]$$

$$= 2023f(2023) - [2022] + \left(\frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{2023} \right)$$

$$= 2023f(2023) - 2022 + f(2023) - 1$$

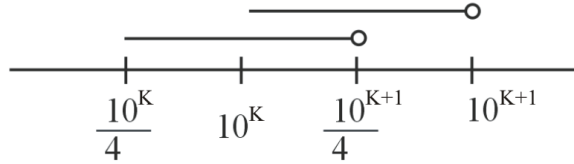
$$= 2024f(2023) - 2023$$

4. Ans (B,D)

$$K \leq \log_{10} 4x < K+1$$

$$10^K \leq 4x < 10^{K+1}$$

$$\frac{10^K}{4} \leq x < \frac{10^{K+1}}{4}$$



$$10^K \leq x < 10^{K+1}$$

5. Ans (B,C)

$$\begin{aligned} S_n &= \sum \left(\frac{4n^4}{4n^2-1} \right) = \sum \frac{1}{4} \left[(4n^2+1) + \frac{1}{4n^2-1} \right] \\ &= \frac{1}{4} \sum 4n^2 + 1 + \frac{1}{4} \sum \frac{1}{(2n-1)(2n+1)} \\ &= 4 + \frac{4}{33} \end{aligned}$$

6. Ans (A,C,D)

$$(0, 1) \rightarrow (2, 0) \text{ OR } (2, 1) \rightarrow (0, 0)$$

$$c = 1$$

$$c = 0$$

$$4a + 2b + c = 0$$

$$4a + 2b + c = 1$$

$$4a + 2b = -1$$

$$2a + b = \frac{1}{2}$$

$$2a + b = \frac{-1}{2}$$

$$\Rightarrow |2a + b| = \frac{1}{2}$$

$$\text{When } c = 1$$

Sum of roots

$$\frac{-b}{a} = \frac{2a + \frac{1}{2}}{a} = \left(P + \frac{1}{2} \right)^2 + \frac{11}{4}$$

$$\Rightarrow 2 + \frac{1}{2a} \geq \frac{11}{4}$$

$$\Rightarrow \frac{1}{2a} \geq \frac{3}{4}$$

$$\frac{1}{a} \geq \frac{3}{2} \Rightarrow 0 < a \leq \frac{2}{3}$$

$$\text{When } c = 0$$

Sum of roots

$$\frac{-b}{a} = \frac{2a - \frac{1}{2}}{a} \geq \frac{11}{4}$$

$$\Rightarrow 2 - \frac{1}{2a} \geq \frac{11}{4}$$

$$\Rightarrow \frac{-1}{2a} \geq \frac{3}{4} \Rightarrow \frac{-1}{a} \geq \frac{3}{2} \Rightarrow \frac{1}{a} \geq \frac{-3}{2}$$

$$\Rightarrow a \geq \frac{-2}{3} \Rightarrow |a| \leq \frac{2}{3}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (D)

$$* A = 2^{\log_3 e^{\log \sqrt{e^3}}} = 2^{\log_3 e^{2 \log e^3}} = 2^{\log_3 3^2} = 2^2$$

$$\Rightarrow A = 4$$

$$* B = \sqrt{17 + 12\sqrt{2}} - \sqrt{17 - 12\sqrt{2}}$$

$$= \sqrt{3^2 + (2\sqrt{2})^2 + 2 \cdot 3 \cdot 2\sqrt{2}} - \sqrt{3^2 + (2\sqrt{2})^2 - 2 \cdot 3 \cdot 2\sqrt{2}}$$

$$= \sqrt{(3 + 2\sqrt{2})^2} - \sqrt{(3 - 2\sqrt{2})^2}$$

$$\Rightarrow B = (3 + 2\sqrt{2}) - (3 - 2\sqrt{2})$$

$$\Rightarrow B = 4\sqrt{2}$$

$$* (\sqrt{2} - 1)^{-C} = \frac{4}{A+B}$$

$$\Rightarrow (\sqrt{2} - 1)^{-C} = \frac{4}{4 + 4\sqrt{2}} = \frac{1}{\sqrt{2} + 1} = (\sqrt{2} - 1)$$

$$\Rightarrow -C = 1 \Rightarrow C = -1$$

$$* 4\alpha = A + B \Rightarrow 4\alpha = 4 + 4\sqrt{2}$$

$$\Rightarrow \alpha = 1 + \sqrt{2}$$

$$\Rightarrow (\alpha - 1)^2 = (\sqrt{2})^2 \Rightarrow \alpha^2 - 2\alpha - 1 = 0$$

Dividing $\alpha^3 + \alpha^2 - 7\alpha + 8$ by $\alpha^2 - 2\alpha - 1$ we will have $(\alpha^3 + \alpha^2 - 7\alpha + 8)$

$$= (\alpha^2 - 2\alpha - 1)(\alpha + 3) + 11 = 0 + 11 = 11$$

8. Ans (C)

$$f(x) = 4x^2 + 8x + t$$

$$f(2) = 16 + 16 + t < 0$$

$$t < -32$$

$$|t| > 32$$

9. Ans (B)

$$g(x) = (x+1)f(x) - x$$

$$= Kx(x-1) \dots (x-n)$$

$$\text{For } n = 10$$

$$g(x) = Kx(x-1) \dots (x-10)$$

$$(x+1)f(x) - x = Kx(x-1) \dots (x-10)$$

$$\text{Put } x = -1$$

$$0 + 1 = K(-1)(-2) \dots (-11)$$

$$K = -\frac{1}{11!}$$

$$f(x) = \frac{-\frac{x(x-1) \dots (x-10)}{11!} + x}{x+1}$$

$$f(11) = \frac{10}{12}$$

10. Ans (D)

$$g(x) = (x+1)f(x) - x$$

$$= Kx(x-1) \dots (x-n)$$

$$\text{For } n = 15$$

$$g(x) = Kx(x-1) \dots (x-15)$$

$$\text{Put } x = -1$$

$$0 + 1 = K(-1)(-2) \dots (-16)$$

$$f(16) = \frac{1+16}{16+1} = 1$$

PART-3 : MATHEMATICS
SECTION-II
1. Ans (0.97)

$$x^3 + px^2 + qx + r = 0 \begin{cases} A \\ G \\ H \end{cases}$$

$$A, G^2, H \text{ are in G.P.}$$

$$(G^2)^2 = AH \Rightarrow G = 0, 1, -1$$

$$G = 1 \Rightarrow AH = 1 \Rightarrow H = \frac{1}{A}$$

$$\text{product of roots} = -r$$

$$A.1. \frac{1}{A} = -r \Rightarrow r = -1$$

$$\text{sum of roots} = -p$$

$$A + 1 + \frac{1}{A} = -p \Rightarrow -p \geq 2 + 1$$

$$p \leq -3$$

$$q = A + \frac{1}{A} + 1 \Rightarrow q \geq 3$$

$$p + q = 0$$

$$\text{possible values of } q \text{ are } 3, 4, 5, \dots, 99$$

$$\text{Total} = 97 \text{ \& } r = \text{fixed} = -1$$

2. Ans (2.00)

$$\sum_{r=1}^{2n} (r(r+1))^2 (-1)^r$$

$$= -\underbrace{(1.2)^2 + (2.3)^2}_{\text{terms}} - \underbrace{(3.4)^2 + (4.5)^2}_{\text{terms}} - \dots - 2n$$

terms

(here a means product)

$$= 2^2[2.4] + 4^2[2.8] + 6^2[2.12] \dots n \text{ terms}$$

$$= 2^3[1^2.4 + 2^2.8 + 3^2.12 + \dots n \text{ terms}]$$

$$= 2^5[1^3 + 2^3 + 3^2 + \dots n \text{ terms}]$$

$$= 2^5 \left(\frac{n(n+1)}{2} \right)^2$$

similarly solve numerator

3. Ans (150.00)

$$T_r = \frac{r(r+1)}{2},$$

$$T_r - 1 = \frac{r(r+1)}{2} - 1 = \frac{r^2 + r - 2}{2} = \frac{(r-1)(r+2)}{2}$$

$$\frac{T_r}{T_r - 1} = \frac{r(r+1)}{(r-1)(r+2)}$$

$$M_{50} = \frac{T_2}{T_2 - 1} \times \frac{T_3}{T_3 - 1} \times \frac{T_4}{T_4 - 1}$$

$$\times \dots \times \frac{T_{48}}{T_{48} - 1} \times \frac{T_{49}}{T_{49} - 1} \times \frac{T_{50}}{T_{50} - 1}$$

$$= \frac{2 \times 3}{1 \times 4} \times \frac{3 \times 4}{2 \times 5} \times \frac{4 \times 5}{3 \times 6} \times \frac{5 \times 6}{4 \times 7}$$

$$\times \dots \times \frac{48 \times 49}{47 \times 50} \times \frac{49 \times 50}{48 \times 51} \times \frac{50 \times 51}{49 \times 52}$$

$$= \frac{2 \times 3^2 \times 4^2 \times 5^2 \times \dots \times 50^2 \times 51}{1 \times 2 \times 3 \times 4^2 \times 5^2 \times \dots \times 49^2 \times 50 \times 51 \times 52}$$

$$= \frac{3 \times 50}{52}$$

$$\text{So } 52 M_{50} = 150$$

4. **Ans (5050.00)**

$$a_1 + a_2 + \dots + a_n - 2\sqrt{a_1} - 2\sqrt{a_2 - 1} \\ \dots - 2\sqrt{a_n - (n-1)} - \frac{n(n-3)}{2} = 0$$

$$(\sqrt{a_1} - 1)^2 + (\sqrt{a_2 - 1} - 1)^2 \\ + \dots + (\sqrt{a_n - (n-1)} - 1)^2 = 0$$

$$a_1 = 1, a_2 = 2, \dots, a_n = n$$

$$\sum_{i=1}^{100} a_i = \frac{100 \times 101}{2} = 5050$$

5. **Ans (8.00)**

$$xy + 2z = \left(\log_2 3 - 4 \log_{\left(\ln \frac{5}{2}\right)} 3 \right) 2 \log_3 \left(\ln \frac{5}{4} \right)$$

$$+ 2 \log_2 \left(\log_{\frac{5}{4}} e \right)$$

$$= 2 \log_2 \left(\ln \frac{5}{4} \right) - 8 - 2 \log_2 \left(\ln \frac{5}{4} \right) = -8$$

$$\therefore |xy + 2z| = 8$$

6. **Ans (17.00)**

$$\text{Let } z = x^2 + 2xy + 3y^2 - 6x - 2y$$

$$x^2 + 2xy + 3y^2 - 6x - 2y - z = 0$$

$$x^2 + 2x(y-3) + 3y^2 - 2y - z = 0$$

$$\because x \in \mathbb{R} \quad D \geq 0$$

$$4(y-3)^2 - 4(3y^2 - 2y - z) \geq 0$$

$$y^2 - 6y + 9 - 3y^2 + 2y + z \geq 0$$

$$-2y^2 - 4y + 9 + z \geq 0$$

$$z \geq 2(y^2 + 2y) - 9$$

$$z \geq 2(y+1)^2 - 11$$

$$z_{\min} = -11$$

$$2 + 3 + 5 + 7 = 17$$

7. **Ans (8.00)**

$$f(x) = \left(x^2 + \frac{a}{2}x \right)^2 + \left(1 + \frac{b}{2}x \right)^2$$

$$+ \frac{1}{4} (8 - a^2 - b^2) x^2 \geq 0 \quad \forall x \in \mathbb{R}$$

$$f(x) \text{ is strictly positive unless } a^2 + b^2 \geq 8$$

8. **Ans (8.50)**

$$\text{Given equation can be written as } x^2 - 3 = 3$$

$$\left[\sin \left(x - \frac{\pi}{6} \right) \right]$$

$$\text{Hence, } x^2 - 3 = -3, 0, 3$$

$$\Rightarrow x = 0, x = \sqrt{3} \text{ is only solution}$$



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ANSWER KEY

PAPER-2 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,D	A,B,C	A,C,D	A,B	A,C,D	B,C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	D	A	B	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	0.60	0.74	533.33	6.02	351.51	2.23 to 2.24	0.56	0.63 to 0.64

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C	A,D	B,C,D	B,C	A,C,D	A		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	C	B	A	D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	1.00	2.00	6.00	2.00	5.00	4.00	4.00	3.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C	B,C	B,C	A,C,D	C,D	C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	B	C	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	9.00	1024.00	2.00	2.00	99.00	3.00	99.00	8.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,B,D)

$$Q = CV = \frac{\epsilon_0 A}{d} V$$

$$\frac{dQ}{Q} = \frac{dA}{A} + \frac{dd}{d} + \frac{dV}{V} \approx 0.045$$

$$\frac{dE}{E} = \frac{dA}{A} + \frac{dd}{d} + \frac{2dV}{V} \approx 0.055$$

$$F = \frac{Q^2}{2\epsilon_0 A}$$

$$\frac{dF}{F} = \frac{dA}{A} + \frac{2dd}{d} + \frac{2dV}{V}$$

When dielectric is inserted

relative error remains same but absolute value

increases.

2. Ans (A,B,C)

	Object needle	u	Lens	u	Image needle
Convex lens	37.1 cm	-13.5 cm	50.6 cm	+27.0 cm	77.6 cm
Convex and Concave lens	37.1 cm	-13.5 cm	50.6 cm	+40.5 cm	91.1 cm

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f_{\text{convex}}} \Rightarrow \frac{1}{27} + \frac{2}{27} = \frac{1}{f} \Rightarrow f_{\text{convex}} = 9$$

$$\frac{1}{v'} - \frac{1}{u'} = \frac{1}{f_{\text{convex}}} + \frac{1}{f_{\text{concave}}} \Rightarrow \frac{2}{27} + \frac{2}{81} = \frac{1}{f_{\text{combo}}}$$

$$\frac{1}{f_{\text{concave}}} = \frac{8}{81} - \frac{1}{9} = -\frac{1}{81} \Rightarrow f_{\text{concave}} = -27 \text{ cm}$$

$$\frac{df}{f^2} = \frac{du}{u^2} + \frac{dv}{v^2} = \frac{0.2}{729} + \frac{0.2 \times 4}{729} = \frac{1.0}{729}$$

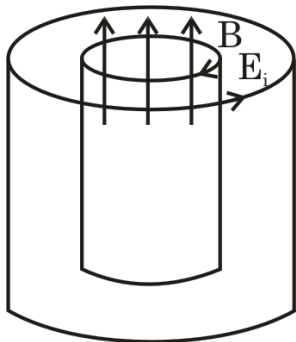
$$df \approx 0.11 \text{ cm}$$

$$\frac{df}{f^2} = \frac{du'}{u'^2} + \frac{dv'}{v'^2} + \frac{du}{u^2} + \frac{dv}{v^2}$$

$$= \frac{0.2}{81^2} \times 4 + \frac{0.2 \times 4}{27^2} + \frac{0.2 \times 4}{27^2} + \frac{0.2 \times 4}{27^2} + \frac{0.2}{27^2}$$

$$df = \frac{17}{81^2} \times 27^2 = \frac{17}{9} \text{ cm}$$

3. Ans (A,C,D)



$$\vec{s} = \frac{\vec{E} \times \vec{B}}{\mu_0}$$

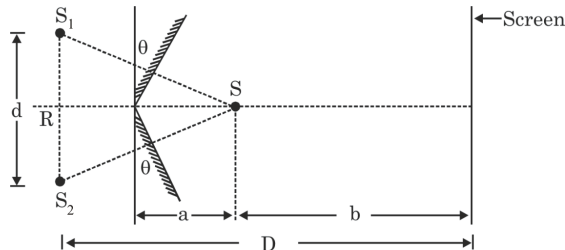
$$E = \frac{r}{2} \frac{dB}{dt} = \mu_0 n i_0 \frac{\omega r}{2} \cos \omega t$$

$$B = \mu_0 n i_0 \sin \omega t$$

$$\vec{E} \times \vec{B} \text{ is inwards}$$

$$\frac{dU}{dt} = \frac{\mu_0 n^2 i_0^2 \omega r}{4} \sin 2\omega t \times 2\pi r \ell$$

4. Ans (A,B)



Distance of S from each mirror = $a \cos \theta$.

$$\therefore SS_1 = SS_2 = 2a \cos \theta$$

$$\therefore S_1 S_2 = d = SS_1 \sin \theta + SS_2 \sin \theta$$

$$= 4a \cos \theta \sin \theta$$

Since θ is small

$$\cos \theta = 1 \text{ and } \sin \theta = \theta$$

$$d = 4a\theta$$

$$\text{distance RS} = SS_1 \cos \theta = 2a \cos^2 \theta = 2a$$

$$\therefore D = RO = RS + SO = (2a + b)$$

$$\therefore \text{Fringe width } \beta = \frac{D\lambda}{d} = \frac{(2a + b)\lambda}{4a\theta}$$

(A) For blue light $\lambda = 4000 \text{ \AA}$, $a = 1 \text{ cm}$, $b = 38 \text{ cm}$

$$\beta_1 = \frac{2 \times 1 + 38}{4 \times 1 \times 0.05} \times 4000 \times 10^{-10} \text{ m}$$

$$= \frac{40}{20} \times 4 \times 10^{-5} \text{ m} = 80 \mu \text{m}$$

(B) Fringe width for light of wavelength $5600 \text{ \AA}$ is

$$\beta_2 = \frac{2 \times 1 + 38}{4 \times 1 \times 0.05} \times 5600 \times 10^{-10} \text{ m} = 112 \mu \text{m}$$

A black line is formed at the position where dark fringes are formed for both the wavelengths. Let the first black line be at a distance y from the central fringe. Let this be the position of m^{th} dark fringe of $4000 \text{ \AA}$ and n^{th} dark fringe of $5600 \text{ \AA}$ light.

$$\therefore y = \left(m - \frac{1}{2}\right) \beta_1 = \left(n - \frac{1}{2}\right) \beta_2$$

$$\Rightarrow \frac{2m-1}{2n-1} = \frac{\beta_2}{\beta_1}$$

For first black line, y should be minimum possible which corresponds to least possible integral values of m and n .

$$\therefore \frac{2m-1}{2n-1} = \frac{7}{5}$$

or, $m = 4, n = 3$

$$\therefore \text{position of first black line } y = \left(m - \frac{1}{2}\right) \beta_1 = 280 \mu$$

(C) Since the interference pattern is symmetric about the central fringe, there will be one dark line at a distance y above it and the other will be y below it.

$$\therefore \text{Relevant answer is } 2y = 560 \mu \text{m.}$$

6. **Ans (B,C,D)**

For destructive interference of

$$\lambda = 552 \text{ nm}$$

$$\Delta x = 2\mu t = (2n + 1) \lambda/2$$

$$t = (2n + 1) \frac{552 \text{ nm}}{4 \times 1.38}$$

$$t = (2n + 1) 100 \text{ nm}$$

$$t = 100, 300, 500, 700 \dots\dots$$

PART-1 : PHYSICS

SECTION-I (ii)

7. **Ans (D)**

$$T = 2\pi\sqrt{\frac{\ell}{g}} \Rightarrow g = \frac{4\pi^2\ell}{T^2}$$

$$\frac{\Delta g}{g} = \frac{\Delta \ell}{\ell} + 2 \frac{\Delta T}{T} \left[\frac{\Delta T}{T} = \frac{\Delta t/n}{t/n} = \frac{\Delta t}{t} \right]$$

$$\frac{\Delta g}{g} = \frac{\Delta \ell}{\ell} + 2 \left(\frac{\Delta t}{t} \right)$$

8. **Ans (A)**

For the pressure waves

$$p_0 = Bks_0$$

$$k = \pi$$

$$s_0 = 2$$

$p_0 = 2\pi B$ and 2 plain waves can be written as

$$p_x = p_0 \cos(340\pi t - \pi x) \text{ and } p_y = -p_0 \cos(\pi y - 340\pi t)$$

by super position principle

$$p = p_x + p_y = 2p_0 \cos\left(\frac{\pi(y-x)}{2}\right) \cos\left(340\pi t - \pi \frac{x+y}{2}\right)$$

$$\Rightarrow \text{pressure amplitude} = 2p_0 \cos\left(\frac{\pi(y-x)}{2}\right)$$

At point (0, 0) amplitude = 0 hence intensity will be minimum.

At point (2, 1) amplitude = $2p_0$ hence intensity will be maximum

$$\text{At point (0.5, 2)} \quad S_x = -2 \cos 340\pi t$$

$$S_y = 2 \sin 340\pi t$$

Path will be circle

PART-1 : PHYSICS

SECTION-II

2. **Ans (0.74)**

$$\text{Zero error} = -\frac{40}{200} \times 0.2 \text{ mm}$$

$$\text{Zero correction} = +\frac{40}{200} \times 0.2 \text{ mm} = 0.04 \text{ mm}$$

$$\text{Reading} = 0.6 \text{ mm} + \frac{100}{200} \times 0.2 \text{ mm} = 0.7 \text{ mm}$$

8. **Ans (0.63 to 0.64)**

$$A = A_0 \left(\frac{\sin \beta}{\beta} \right)$$

$$\text{or } \frac{A}{A_0} = \frac{\sin \beta}{\beta}$$

$$\text{Now } \beta = \frac{\pi a \sin \theta}{\lambda} = \frac{\pi}{2}$$

$$\therefore \frac{A}{A_0} = \frac{\sin\left(\frac{\pi}{2}\right)}{\left(\frac{\pi}{2}\right)} = \frac{2}{\pi}$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (A,B,C)**

$$(x - k)(x - 1) + 1 = 0$$

$$x^2 - (k + 1)x + k + 1 = 0$$

$$D = (k + 1)^2 - 4(k + 1)$$

$$D = k^2 - 2k + 1 - 4$$

$$D = (k - 1)^2 - 4$$

for D to be a perfect square

$$(k - 1)^2 = 4$$

$$\Rightarrow k - 1 = \pm 2$$

$$D = k^2 - 2k + 1 - 4$$

$$D = (k - 1)^2 - 4$$

For D to be a perfect square

$$(k - 1)^2 = 4$$

$$\Rightarrow k - 1 = \pm 2$$

$$\Rightarrow k = -1 \text{ or } 3$$

$$\log_2(a^2 + 1)^3 = -1 \text{ or } 3$$

$$(a^2 + 1)^3 = \frac{1}{2} \text{ or } 2^3$$

$$a^2 + 1 = 2$$

$$\Rightarrow a^2 = 1$$

$$\Rightarrow a = 1 \text{ or } -1$$

2. Ans (B,C)

$$x^2 + x \in \left[-\frac{1}{4}, \infty\right)$$

$$\text{Put } x^2 + x = t$$

$$t^2 + at + 4 = 0 \text{ doesn't}$$

$$\text{have real root in } \left[-\frac{1}{4}, \infty\right)$$

Case-I 

$$\Rightarrow a \in (-\infty, -4] \cup [4, \infty)$$

$$-\frac{b}{2a} < -\frac{1}{4} \Rightarrow -\frac{a}{2} < -\frac{1}{4} \Rightarrow a > \frac{1}{2}$$

$$f\left(-\frac{1}{4}\right) > 0$$

$$\frac{1}{16} - \frac{a}{4} + 4 > 0$$

$$\frac{a}{4} < \frac{65}{16}$$

$$a < \frac{65}{4}$$

$$a \in \left[4, \frac{65}{4}\right)$$

Case-II 

$$D < 0 \Rightarrow a \in (-4, 4)$$

$$\text{Thus final Ans. } a \in \left(-4, \frac{65}{4}\right)$$

3. Ans (B,C)

$$a^3 - 8b^3 + 64 = -24ab$$

$$a^3 + (-2b)^3 + (4)^3 = 3(a)(-2b)(4)$$

$$\Rightarrow 4 - 2b + a = 0$$

$$\Rightarrow x = -2 \text{ is a root}$$

$$\alpha + \beta = -b$$

$$0 < b < 3$$

$$0 > -b > -3$$

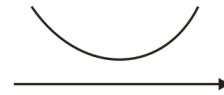
$$0 > \alpha + \beta > -3$$

$$0 > -2 + \beta > -3$$

$$2 > \beta > -1$$

4. Ans (A,C,D)

$$f(0) = 1$$



quadratic must be always positive

$$\Rightarrow \text{coefficient of } x^2 > 0$$

$$\log_{(3a)}^{(a^2-4)} > 0$$

Case-I

$$a^2 - 4 > 1 \text{ and } 3a > 1$$

$$a^2 - 5 > 0 \text{ and } a > \frac{1}{3}$$

$$a \in (-\infty, -\sqrt{5}) \cup (\sqrt{5}, \infty) \text{ and } a > \frac{1}{3}$$

$$\Rightarrow a \in (\sqrt{5}, \infty)$$

Case-II

$$0 < a^2 - 4 < 1 \quad 3a < 0$$

$$1 < a^2 < 5 \quad a < 0$$

$$a \in (-\sqrt{5}, -1) \cup (1, \sqrt{5})$$

$$\Rightarrow a \in (-\sqrt{5}, -1)$$

$$\text{Final Ans. } a \in (-\sqrt{5}, -1) \cup (\sqrt{5}, \infty)$$

5. Ans (C,D)

$$(e^{4x} + 1) - a(e^{3x} + e^x) + 11e^{2x} = 0$$

$$(e^{2x} + e^{-2x}) - a(e^x + e^{-x}) + 11 = 0$$

$$\text{Put } e^x + e^{-x} = t$$

$$t^2 - at + 9 = 0$$

For x to have 2 distinct positive value t must have two roots greater than 2.



$$D > 0 \quad -\frac{b}{2a} > 2 \quad f(2) > 0$$

$$a^2 - 36 > 0 \quad \frac{a}{2} > 2 \quad 4 - 2a + 9 > 0$$

$$a \in (-\infty, -6) \cup (6, \infty) \quad a > 4 \quad 2a < 13$$

$$a < \frac{13}{2}$$

$$a \in \left(6, \frac{13}{2}\right)$$

6. Ans (C,D)

$$T_n = S_n - S_{n-1} = nnnnn$$

$$\Rightarrow T_n = n [11111.....]$$

$$= n [1 + 10 + 100 + \dots n \text{ terms}]$$

$$T_n = \frac{n(10^n - 1)}{10 - 1}$$

PART-3 : MATHEMATICS
SECTION-I (ii)
7. Ans (A)

$$\log_3 2 = a, b = \log_3 5 / \log_3 4 = \frac{\log_3 5}{2 \log_3 2}$$

$$\Rightarrow \log_3 2 = a \Rightarrow \log_3 5 = 2ab$$

$$c = \frac{\log_3 6}{\log_3 5} = \frac{\log_3 2 + 1}{\log_3 5} = \frac{a + 1}{2ab}$$

$$\Rightarrow 2bc = \frac{a + 1}{a} \Rightarrow (3) = (4)$$

$$\text{Now (P) } \log_{10} 135$$

$$= \frac{\log_3 (3^3 \times 5)}{\log_3 2 \times 5} = \frac{3 + \log_3 5}{a + 2ab} = \frac{3 + 2ab}{a(1 + 2b)}$$

$$(Q) \log_4 36 = 1 + \log_4 9$$

$$= 1 + \log_2 3 = 1 + \frac{1}{a} = \frac{a + 1}{a}$$

$$(R) \log_{24} 12 = \frac{\log_3 12}{\log_3 24} = \frac{1 + 2 \log_3 2}{1 + 3 \log_3 2}$$

$$= \frac{1 + 2a}{1 + 3a} = \frac{1/a + 2}{1/a + 3} = \frac{2bc - 1 + 2}{2bc - 1 + 3}$$

$$\left(\because \frac{a + 1}{2} = 2bc \right)$$

$$= \frac{2bc + 1}{2bc + 2}$$

$$(S) 2 \log 3 + \frac{1 + \log_5 3}{1 + \log_5 2} = 2 \log 3$$

$$+ \frac{\log_3 5 + \log_5 3}{\log_5 3 + \log_5 2} = 2 \log 3 + \frac{\log_5 15}{\log_5 10}$$

$$= 2 \log 3 + \log 15 = \log 135$$

8. Ans (B)

$$(P) a, b, c \text{ in A.P. (1)}$$

$$x, y, z \text{ in G.P. (2)}$$

$$\log x, \log y, \log z \text{ in A.P. (3)}$$

$$y, z, u \text{ in G.P. (4)}$$

$$3 \log y, 3 \log z, 3 \log u \text{ in A.P. ... (5)}$$

$$\log y^3, \log z^3, \log u^3 \text{ in A.P. ... (6)}$$

$$\text{Now from (1) - (3) + (6)}$$

$$a - \log \frac{x}{y^3}, b - \log \frac{y}{z^3}, c - \log \frac{z}{u^3} \text{ in A.P.}$$

$$(Q) x, y, z, u \text{ in G.P.}$$

$$\Rightarrow xy, yz, zu \text{ in G.P.}$$

$$\Rightarrow kxy, kyz, kzu \text{ in G.P.}$$

where $k = b - a = c - b = d - c = \text{common difference of A.P.}$

$$(R) \text{ If } \frac{a(b+c)}{bc}, \frac{b(c+a)}{ca}, \frac{c(a+b)}{ab} \text{ in A.P.}$$

$$\Rightarrow \frac{a(b+c)}{bc} + 1, \frac{b(c+a)}{ca} + 1, \frac{c(a+b)}{ab} + 1 \text{ in A.P.}$$

$\div (ab + bc + ca)$ and multiply by a, b, c

$$\Rightarrow a, b, c \text{ in A.P. (which is given)}$$

$$(S) (a^2 - b^2), (b^2 - c^2), (c^2 - d^2) \text{ in A.P.}$$

$$(1)$$

$$(x^2 - y^2), (y^2 - z^2), (z^2 - u^2) \text{ in G.P. (2)}$$

from (1) and (2) result will be an A.G.P.

9. Ans (C)

$$(Q) \text{ If } (a^2 + b^2 + c^2)(b^2 + c^2 + d^2) < (ab + bc + cd)^2$$

then a, b, c in G.P.

and if $c(a - b) = a(b - c) \Rightarrow a, b, c$ in H.P.

$\Rightarrow$ then a, b, c, d all are equal.

$$(S) 2a_2 = a_1 + a_3 \text{(i)}$$

$$a_3^2 = a_2 a_4 \text{(ii)}$$

$$a_4 = \frac{2a_3 a_5}{a_3 + a_5} \text{(iii)}$$

from (i), (ii) & (iii) $\Rightarrow a_3^2 = a_1 a_5$.

10. Ans (A)

(P) $\log\left(\frac{2}{3}|x-2|\right)^2 \geq 0$ if $2 \leq \frac{2}{3}|x-2|$ lies on same side of unity

$$\Rightarrow \frac{2}{3}|x-2| > 1 \Rightarrow x-2 > \frac{3}{2} \quad \text{or} \quad x-2 < -\frac{3}{2}$$

$$\therefore x \in \left(-\infty, \frac{1}{2}\right) \cup \left(\frac{7}{2}, \infty\right)$$

(Q) $D > 0$

$$\Rightarrow (\alpha - \beta)^2 - 4(1 - \alpha - \beta) > 0$$

$$\text{or } \beta^2 + \beta(4 - 2\alpha) + (\alpha^2 - 4 + 4\alpha) > 0$$

This is true for all real β

$$\Rightarrow 4(2 - \alpha)^2 - 4(\alpha^2 - 4 + 4\alpha) < 0$$

$$\text{or } -8\alpha + 8 < 0 \quad \text{or } \alpha > 1.$$

(R) Let the roots of $x^3 - 6x^2 + 2ax - 4b = 0$ be

$$\alpha - d, \alpha, \alpha + d$$

$$\Rightarrow (\alpha - d) + \alpha + (\alpha + d) = 6,$$

$$\alpha(\alpha - d) + \alpha(\alpha + d) + (\alpha - d)(\alpha + d) = 2a$$

$$\& \alpha(\alpha - d)(\alpha + d) = 4b$$

$$\therefore \alpha = 2$$

$$\text{hence } 2a = 12 - d^2 \& 2b = 4 - d^2$$

But $a \in (5, 6)$

$$\Rightarrow 0 < d^2 < 2$$

$$\therefore 1 < b < 2$$

(S) $D \geq 0$

$$36\lambda^2 - 4(2 - 2\lambda + 9\lambda^2) \geq 0$$

$$2 - 2\lambda \leq 0$$

$$\lambda \geq 1$$

$$af(3) > 0$$

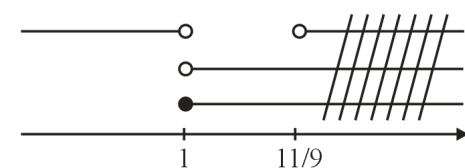
$$-\frac{b}{2a} > 3 \quad 9 - 18\lambda + 2 - 2\lambda + 9\lambda^2 > 0$$

$$3\lambda > 3 \quad 9\lambda^2 - 20\lambda + 11 > 0$$

$$\lambda > 1 \quad (\lambda - 1)(9\lambda - 11) > 0$$

$$\lambda \in (-\infty, 1) \cup \left(\frac{11}{9}, \infty\right)$$

Intersection



$$\lambda \in \left(\frac{11}{9}, \infty\right)$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (9.00)

Given quadratic is

$$(m^2 + 1)x^2 - 3x + (m^2 + 1)^2 = 0$$

$$\therefore \text{sum} = \frac{3}{(m^2 + 1)}$$

$$\therefore (\text{sum})_{\text{maximum}} = 3 \text{ (for } m = 0)$$

$$\text{product} = (m^2 + 1)$$

$$(\text{product})_{\text{minimum}} = 1 \text{ (for } m = 0)$$

$$(\text{sum})_{\text{max}} = q \text{ and } (\text{product})_{\text{min}} = p \text{ (Given)}$$

the first term of G.P. = $p + 2$

$$= 1 + 2 = 3.$$

$$\text{and common ratio} = \frac{2}{q} = \frac{2}{3}$$

$$\therefore S_{\infty} = \frac{3}{1 - \frac{2}{3}} = 9$$

2. Ans (1024.00)

$$S = \frac{x}{1-x^2} + \frac{x^2}{1-x^4} + \frac{x^4}{1-x^8} + \dots 10 \text{ terms}$$

$$S = \frac{1}{1-x} - \frac{1}{1-x} + \frac{x}{1-x^2} + \frac{x^2}{1-x^2} + \dots$$

$$= \frac{1}{1-x} + \frac{1}{1-x} \left[\frac{x}{1+x} - 1 \right] + \frac{x^2}{1-44}$$

$$= \frac{1}{1-x} - \frac{1}{1-x^2} + \frac{x^2}{1-x^4} + \dots$$

$$= \frac{1}{1-x} - \frac{1}{1-x^{2^{10}}}$$

$$= \frac{1}{1-x} - \frac{1}{1-x^{1024}}$$

$$\therefore k = 1024$$

3. Ans (2.00)

$$\sum_{k=1}^{\infty} \frac{6^k}{(3^k - 2^k)(3^{k+1} - 2^{k+1})}$$

$$= \sum \frac{3^k}{(3^k - 2^k)} - \frac{3^{k+1}}{(3^{k+1} - 2^{k+1})}$$

$$= \left(\frac{3}{3-2} - \frac{3^2}{3^2-2^2} \right) + \left(\frac{3^2}{3^2-2^2} - \frac{3^3}{3^3-2^3} \right)$$

$$+ \dots \left(\frac{3^k}{3^k-2^k} - \frac{3^{k+1}}{3^{k+1}-2^{k+1}} \right)$$

for $S_{\infty} = S_k = \lim_{k \rightarrow \infty} 3 - \frac{1}{1 - \left(\frac{2}{3}\right)^{k+1}} = 2$

4. Ans (2.00)

$$x^2 + 4kx + 3k^2 > 1 + 2k$$

$$\Rightarrow x^2 + 2kx \leq (3k^2 - 8k + 4)$$

$$\Rightarrow (-\infty, a) \cup (b, \infty) = a + b = ?$$

$$x^2 + 4kx + 3k^2 - 1 - 2k > 0, \text{ on factorising}$$

$$\Rightarrow f(x) = (x + 3k + 1)(x + k - 1) > 0 \quad \dots(1)$$

Let solution of this $(-\infty, \delta) \cup (\gamma, \infty)$

$$\Rightarrow x^2 + 2kx - 3k^2 + 8k - 4 \leq 0$$

factorising this equation

$$(x + 3k - 2)(x - k + 2) \leq 0 \quad \dots(2)$$

$\hookrightarrow [\alpha, \beta]$

Now it has been given that there must be atleast one common solution of equation no (1) & (2)

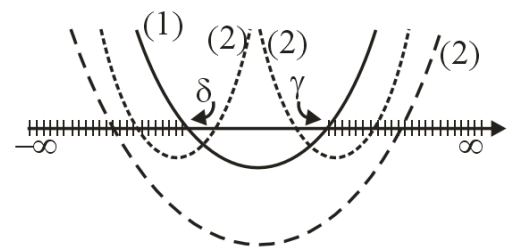
So this kind of possibilities are there

$$f(x) = (x + 3k + 1)(x + k - 1) > 0$$

either :

$$f(-3k + 2) > 0$$

$$\Rightarrow (-3k + 2 + 3k + 1)(-3k + 2 + k - 1) > 0$$



===== Solution of eq. (1)

$$\Rightarrow a + b = \frac{1}{2} + \frac{3}{2} \Rightarrow a + b = 2$$

$$\Rightarrow 3(-2k + 1) > 0 \Rightarrow k < \frac{1}{2}$$

$$\Rightarrow k \in \left(-\infty, \frac{1}{2}\right) \quad \dots(3)$$

$$\text{or } f(k - 2) > 0$$

$$\Rightarrow (k - 2 + 3k + 1)(k - 2 + k - 1) > 0$$

$$\Rightarrow (4k - 1)(2k - 3) > 0$$

$$\Rightarrow k \in \left(-\infty, \frac{1}{4}\right) \cup \left(\frac{3}{2}, \infty\right) \quad \dots(4)$$

so (3) $\cup$ (4)

$$\Rightarrow k \in \left(-\infty, \frac{1}{2}\right) \cup \left(\frac{3}{2}, \infty\right)$$

$$\Rightarrow k \in (-\infty, a) \cup (b, \infty)$$

$$\Rightarrow a = \frac{1}{2}, b = \frac{3}{2}$$

$$\Rightarrow a + b = \frac{1}{2} + \frac{3}{2} \Rightarrow a + b = 2$$

5. Ans (99.00)

Given that, when x^{100} is divided by $x^2 - 3x + 2$, the remainder is $(2^{k+1} - 1)x - 2(2^k - 1)$

$$\text{so } x^{100} = q(x)(x^2 - 3x + 2) + ax + b$$

(where $q(x)$ is quotient and $ax + b$ is remainder also

x^{100} is dividing by quadratic so remainder is in the form of $ax + b$).

By putting $x = 1$,

$$1^{100} = a + b \Rightarrow a + b = 1 \quad \dots(i)$$

By putting $x = 2$,

$$2^{100} = 2a + b \Rightarrow 2a + b = 2^{100} \quad \dots(ii)$$

from (i) and (ii), we get

$$\Rightarrow a = 2^{100} - 1, b = 2 - 2^{100}$$

$$\Rightarrow \text{remainder} = (2^{100} - 1)x + (2 - 2^{100})$$

$$= (2^{99+1} - 1)x - 2(2^{99} - 1)$$

$$\Rightarrow k = 99$$

6. Ans (3.00)

$$\alpha + r = \frac{4}{A}, \quad \alpha r = \frac{1}{A}$$

$$\alpha + r = 4\alpha r$$

$$\text{or } \frac{1}{\alpha} + \frac{1}{r} = 4 \quad \dots(i)$$

$$\text{Again } \beta + \delta = 6\beta\delta$$

$$\text{or } \frac{1}{\beta} + \frac{1}{\delta} = 6 \quad \dots(ii)$$

$\therefore \alpha, \beta, \gamma, \delta$ are in H.P.

$$\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}, \frac{1}{\delta} \quad \text{A.P.}$$

But the no. are $a - 3d, a - d, a + d, a + 3d$

$$4a = 6 + 4 = 10 \text{ or } a = \frac{5}{2}$$

$$\frac{1}{\alpha} + \frac{1}{\gamma} = 4 \quad (\text{Given}) \text{ then}$$

$$a - 3d + a + d = 4$$

$$2a - 2d = 4$$

$$a - d = 2$$

$$d = \frac{1}{2}$$

$$\therefore \alpha = 1, \beta = 2, \gamma = 3, \delta = 4$$

$$\lambda = 3, \lambda + 5 = 8$$

7. Ans (99.00)

$$\frac{1}{a_1}, \frac{1}{a_2}, \dots, \frac{1}{a_{100}} \rightarrow \text{A.P.}$$

$$\frac{1}{a_{i+1}} - \frac{1}{a_i} = d$$

$$\frac{a_i - a_{i+1}}{a_i a_{i+1}} = d$$

$$\Rightarrow a_i a_{i+1} = \frac{a_i - a_{i+1}}{d} \text{ put in Ques.}$$

$$\sum_{i=1}^{99} \frac{a_i - a_{i+1}}{a_i a_{i+1} d} = \frac{(a_1 - a_2) + (a_2 - a_3) + \dots + (a_{99} - a_{100})}{a_1 a_{100} d}$$

$$\frac{a_1 - a_{100}}{a_1 a_{100} d}$$

$$\left(\frac{1}{a_{100}} - \frac{1}{a_1} \right) \cdot \frac{1}{d}$$

$$\frac{99d}{d} = 99$$

8. Ans (8.00)

$$\left(\frac{1}{a} - 1 \right) \left(\frac{1}{b} - 1 \right) \left(\frac{1}{c} - 1 \right)$$

$$= \frac{1}{abc} - \left(\frac{1}{ab} + \frac{1}{bc} + \frac{1}{ca} \right) + \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) - 1$$

$$= \frac{1}{abc} - \left(\frac{a+b+c}{abc} \right) + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} - 1$$

$$= \frac{1}{a} + \frac{1}{b} + \frac{1}{c} - 1$$

Apply AM $\geq$ HM

$$\frac{a+b+c}{3} \geq \frac{3}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}}$$

$$\Rightarrow \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 9$$

$$\Rightarrow \frac{1}{a} + \frac{1}{b} + \frac{1}{c} - 1 \geq 8$$

$$\Rightarrow \text{Min value} = 8$$